S&P GLOBAL HACKATHON · FINANCIAL RISK

Autonomous AI/NLP Financial Risk Engine

From financial-news text to inspectable risk signals and illustrative portfolio scenarios.

Krishna Kapoor
Thapar Institute of Engineering and Technology

News adds context to market and reporting data

Two outputs, one signal layer

TRADITIONAL SYSTEMS

Information latency

News and market data arrive on different timelines; text can add context for an analyst to review.

Tactical index rebalancing

Sentiment-based weight changes in a synthetic 15-stock basket.

OUR APPROACH

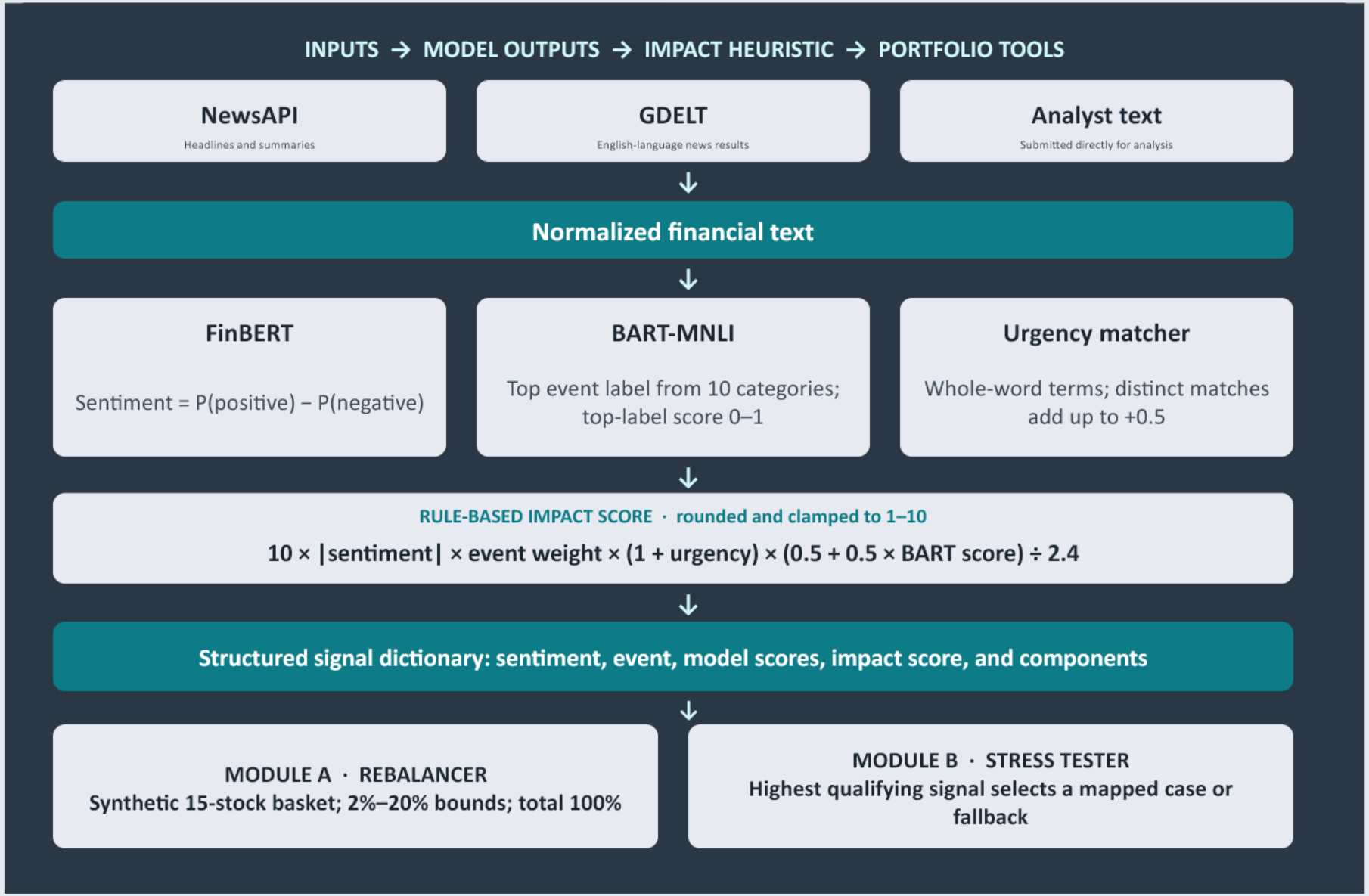
Text becomes signal

The prototype turns financial and geopolitical text into model-backed signals for two synthetic portfolio workflows.

Strategic wholesale portfolio stress testing

Illustrative cross-asset shock estimates for analyst review.

From financial text to analyst review



INPUTS

NewsAPI, GDELT, and analyst text enter the same analysis pipeline.

MODEL OUTPUTS

FinBERT estimates sentiment; BART-MNLI supplies an event label and top-label score.

DECISION ROUTING

FastAPI exposes signal history and portfolio endpoints; Streamlit reads the API for review.

IMPLEMENTATION HIGHLIGHTS

Implementation combines transparent formulas with pretrained models

IMPACT SCORE

Raw Score =
10 × |Sentiment| × W_event
× (1 + B_urgency) ×
(0.5 + 0.5 × BART score)

Impact Score =
max(1, min(10, round(
Raw Score / 2.4)))

Heuristic score, rounded and clamped to 1–10

Sentiment

FinBERT returns P(positive) − P(negative), producing a score from −1.0 to +1.0.

Event taxonomy

BART-MNLI chooses the highest-scoring label from 10 categories. Its top-label score (0–1) feeds the impact formula.

Stack

Transformers/PyTorch, FastAPI/Pydantic, NumPy/Pandas, Streamlit/Plotly, and Pytest.

KEY RESULTS

Risk signals feed two illustrative portfolio workflows

Output	Trigger or control	Decision produced
Risk signal	Each headline is scored by FinBERT and classified by BART-MNLI.	Structured signal includes sentiment, event, model scores, impact score, and components.
Module A	Absolute mean sentiment ≥ threshold (default 0.05; range 0.01–0.50).	Rebalances a synthetic 15-stock basket; weights stay 2%–20% and sum to 100%.
Module B	Impact score ≥7; auto mode selects the highest-impact signal and its event scenario.	Runs one configured case (or fallback) on the synthetic $40.3M portfolio using simplified shock assumptions.

DOMAIN IMPACT

Prototype value and current scope

Analyst triage

Surfaces headline signals for review; price-prediction performance has not been backtested.

Scenario review

Estimates portfolio losses under simplified event shocks and assumptions.

Explainability

Exposes sentiment, event weight, urgency, classifier score, and impact components.

Future regulatory work

CCAR/Basel workflows need separate capital calculations, validation, and controls.

PRODUCTION VALIDATION

Next step: validate the signal-to-decision loop

Calibrate scores against historical market reactions and labeled financial events.

Add entity linking, deduplication, multilingual coverage, and model monitoring for distribution drift.

Connect to governed S&P Global data products and authenticated enterprise feeds.

Backtest rebalancing and stress scenarios with transaction costs, liquidity constraints, and analyst review.

Current assumptions include synthetic portfolios, fixed basket weights, and headline-level text.